Replicating Figure F1 and Equation (1) from Athey, Inoue, and Tsugawa (2025)

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1 The Original Paper

This replication is from the paper *Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance* by Susan Athey, Kosuke Inoue, and Yusuke Tsugawa, published in *AEA Papers and Proceedings* (2025).[1] The paper studies how to optimally assign treatment in settings where a randomized experiment has imperfect compliance. Imperfect compliance is when being assigned to that treatment group does not mean that you will receive the treatment. This creates issues since one cannot simply compare the control and treatment groups directly, due to bias. The central research question is how heterogeneity in both treatment benefit and compliance rates should jointly inform targeting decisions.

The empirical strategy is with regards to the Oregon Health Insurance Experiment (OHIE), in which lottery selection served as a randomized instrument (Z_i) for Medicaid enrollment (W_i). The authors use Generalized Random Forests (GRF) to estimate heterogeneous treatment effects, decomposing the conditional Intention to Treat (ITT) into a compliance component and a benefit component via the Conditional Local Average Treatment Effect (CLATE) formula.

2 The Result of Interest

The replication results are from the equation and table that are below:

[1] Athey, Susan, Kosuke Inoue, and Yusuke Tsugawa. 2025. "Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance." *AEA Papers and Proceedings* 115: 209–214. Supplemental Appendix available at the same URL.

1. **Equation (1)**: the Conditional Local Average Treatment Effect (CLATE), defined as:

$$\tau_{CL}(x) = \frac{\tau_Y(x)}{\tau_W(x)} \tag{1}$$

   where $\tau_Y(x)$ is the conditional Intention to Treat (ITT) and $\tau_W(x)$ is the compliance Conditional Average Treatment Effect (CATE). The paper reports a mean Local Average Treatment Effect (LATE) for debt of approximately 0.15.

2. **Figure F1**: the Targeting Operator Characteristic (TOC) curve for compliance, with a reported Area Under the Targeting Operator Characteristic curve (AUTOC) = 0.0412 (SE = 0.0068). This is the headline measure of how much can be gained by targeting individuals with a high estimated compliance.

These results are central to the paper's argument. The issue of people being assigned to a treatment group, but not everyone taking it is prevalent in random experiments, including but not limited to a Medicaid lottery. Equation 1 focuses on the treatment effect for those that comply by dividing the ITT or outcome CATE (which is the expected health benefit from offering that treatment to those that have observed characteristics which are determined prior to treatment) by the conditional compliance rate (which is the subpopulation of those with observable characteristic who receive treatment if offered). The equation estimates the health benefits of the treatment given that they have observable characteristics prior and receive treatment if offered, meaning they are a complier. Figure F1 focuses on the difference between targeting high compliance individuals rather than offering Medicaid at random. The x-axis measures the percent of people that are targeted (q), and the y-axis measures the Targeting Operator Characteristic (TOC) which compares the targeted group's compliance rate to the average compliance rate. Both this equation and figure are essential for understanding compliance. The equation is intended to show the health benefit of people who are compliers receiving treatment, and the figure shows who should be offered Medicaid, the individuals with high compliance, or random assignment. For these impacts with respect to the objective of the paper, these reults are the ones worth reproducing.

3 Data and Methods

The data come from the Oregon Health Insurance Experiment (OHIE) (Finkelstein et al. 2018),[2] which is a randomized study evaluating the causal effects of Medicaid coverage on

[2] Finkelstein, Amy, Katherine Baicker, Sarah Taubman, et al. 2018. Replication Data for: "Oregon Health Insurance Experiment." Harvard Dataverse. `https://doi.org/10.7910/DVN/SJG1ED`.

the outcome of low-income Oregon adults. There are a total of 12,229 respondents, and of them, there were exclusions among the individuals that were missing data on treatment, age, sex, race, or education, as well as transgender individuals, yielding a final sample of $N = 12{,}208$.

`code/preprocess.R` loads the raw data from `input/`, removes duplicate rows and junk columns, and imputes missing baseline covariates using the `missRanger` random forest package (as specified in Supplemental Appendix A.5). The imputed dataset is saved to `temp/`.

`code/analysis.R` reads the imputed data and estimates two GRF models. An instrumental forest (`instrumental_forest`) is utilized to estimate Equation (1), the CLATE $\tau_{CL}(x)$. A causal forest (`causal_forest`) with outcome W_i and treatment Z_i estimates the compliance CATE $\tau_W(x)$, which is then used to rank individuals for the TOC curve. The AUTOC is computed using `rank_average_treatment_effect` with doubly robust scores, following Yadlowsky et al. (2024).

The 23 pre-treatment covariates follow the paper's Supplemental Appendix A.5: age, sex, race/ethnicity, education, medical conditions (hypertension, diabetes, high cholesterol, asthma, heart attack, CHF, emphysema, kidney failure, cancer, depression), pre-lottery healthcare expenditures and ED utilization, and household size at lottery entry.

4 Replication Results

Table 1 compares our replicated estimates to the paper's reported values. Figure 1 shows the replicated TOC curve for compliance.

Table 1: Replication of Figure F1 AUTOC from Athey, Inoue, and Tsugawa (2025)

	Replicated	Paper
AUTOC	0.0414	0.0412
Standard Error	(0.0087)	(0.0068)
95% CI	[0.0243, 0.0585]	[0.0279, 0.0545]
Mean CLATE	0.1253	≈ 0.15
N	12,208	

The replicated AUTOC of 0.0414 is extremely close to the paper's 0.0412, with a difference of just 0.0002 — is well within sampling variation. The AUTOC tells us the average gain in compliance rate due to targeting rather than random assignments. In either case, it is about a 4.1 percentage points more if the compliance CATE is utilized to target people, rather than random experiments. The mean CLATE of 0.1253 is consistent with the paper's reported overall LATE of approximately 0.15 for debt reduction among compliers. In

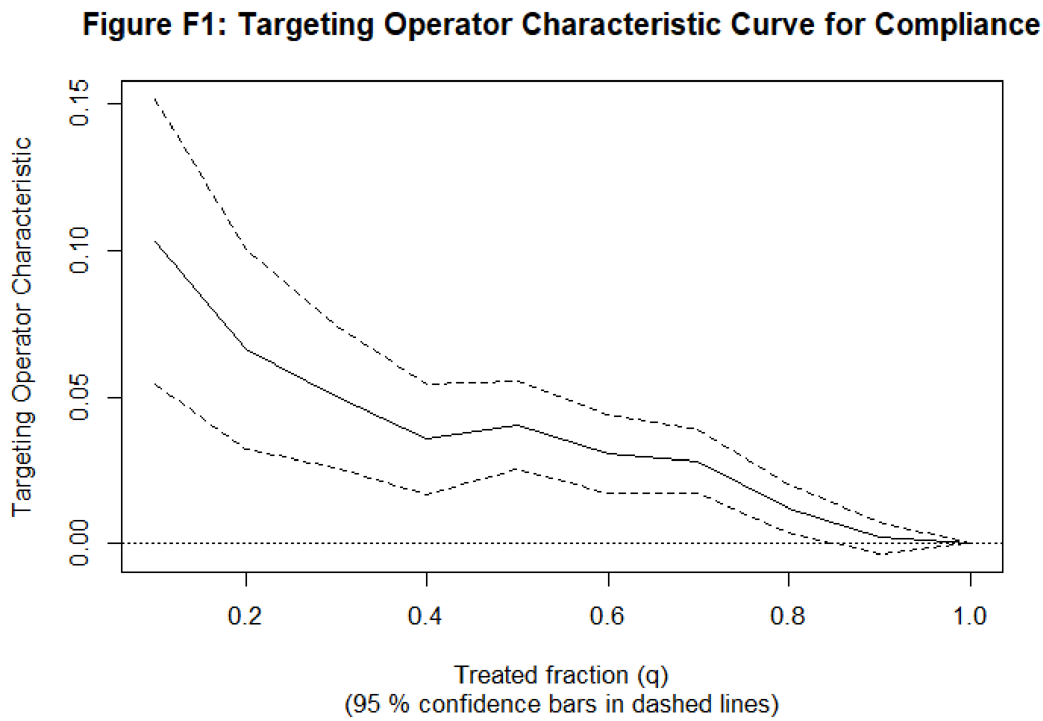

Figure 1: Replication of Figure F1: Targeting Operator Characteristic Curve for Compliance. Outcome: Compliance (W_i); Treatment: Eligibility (Z_i); Ranking rule: $r(\cdot;\ \hat{\tau}_W(X_i))$.

other words, it reveals that Medicaid reduced the probability of obtaining medical debt for compliers by about 12% to 15%. Both results confirm that there is substantial, predictable heterogeneity in compliance rates across individuals. The number of operations (N) is also consistent with the paper's at 12,208, so, there were no observations wrongfully omitted.

The replicated standard error (0.0087) is slightly larger than the paper's (0.0068), likely reflecting minute differences in the `grf` package version or number of bootstrap samples used. The error term is small, so the increase in gain from targeting is more than simply noise.

5 What I Learned

The easiest part of the replication was finding the data to replicate. The paper was available at the American Economic Association, and the data was located by link at that site, and its true location is openICPSR. The access to the data, paper, and any relevant materials such as the Supplemental Appendix made the process of replication much more efficient.

The hardest part of the replication was in fact the data preparation, specifically understanding how missing data should be handled. An initial approach using `na.omit()` reduced the sample from 12,208 to 6,816 by dropping all rows with any missing values. The paper, however, specifies `missRanger` imputation in Supplemental Appendix A.5 — a detail that was essential in order to accurately obtain the results. This shows how critical it is for authors to document preprocessing decisions as carefully as estimation decisions.

A second challenge was the covariate selection. The paper describes covariates in prose rather than providing the exact variable name, this required careful cross-referencing between the dataset and the paper's description. For example, the household size variable (`numhh_list`) is a standard OHIE control but it is not explicitly listed in the paper's covariate description.

Overall, the replication taught me that GRF-based analyses are highly sensitive to sample construction. If you use the wrong command, like mentioned above, you run the risk of losing even a great portion of your sample. So you must feed the forest method correct data. Also that the trees used for the treatment effects are blind to the individual's outcome, and that the predictions are made on data that the tree isn't trained on. The latter two are necessary to ensure that overfitting is not an issue, and the AUTOC standard error is not significant. If I were the original author, I would have provided an exact variable list and the preprocessing script directly in the replication package rather than describing the covariates in prose. This would reduce speculation, and potentially significant errors which in turn would ease the replication process.[3]

References

[1] Athey, Susan, Kosuke Inoue, and Yusuke Tsugawa. 2025. "Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance." *AEA Papers and Proceedings* 115: 209–214.

[2] Finkelstein, Amy, Katherine Baicker, Sarah Taubman, et al. 2018. Replication Data for: "Oregon Health Insurance Experiment." Harvard Dataverse. https://doi.org/10.7910/DVN/SJG1ED.

[3] Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized Random Forests." *Annals of Statistics* 47(2): 1148–1178.

[4] Yadlowsky, Steve, Scott Fleming, Nigam Shah, Emma Brunskill, and Stefan Wager. 2024. "Evaluating Treatment Prioritization Rules via Rank-Weighted Average Treatment Effects." *Journal of the American Statistical Association*.

[3] Claude (Anthropic) was used for assistance with R coding, LaTeX formatting, and Git/GitHub setup in this replication project.